3D Neutrino-Cooled Accretion Disk Dynamo Cycle UQFF

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PAPER_820: 3D Neutrino-Cooled Accretion Disk Dynamo Cycle UQFF

Unified Quantum Field Framework — Whitepaper 820

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Abstract

This paper integrates the 3D neutrino-cooled accretion disk magnetic dynamo cycle into the Quadri-adic UQFF. The 380 ms simulation of a 0.03 $M_\odot$ disk around a 3 $M_\odot$ BH (spin $\chi = 0.8$) reveals self-sustaining dynamo cycles with $\tau_{dynamo} \approx 20$ ms periodicity driven by the MRI (Magneto-Rotational Instability). The electron fraction self-regulates from midplane $Y_e \approx 0.1$ to outflow $Y_e \approx 0.2$ via neutrino absorption, governing the lanthanide content of early kilonova ejecta. Mass ejection $M_{ej} \approx 0.01 M_\odot$ at $v_\infty \approx 0.1c$ aligns with GW170817 red kilonova constraints.

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1. Introduction

The MRI instability drives turbulent angular momentum transport in the accretion disk, with the resulting turbulence generating a self-sustaining dynamo. In the neutrino-cooled regime (relevant for post-merger BH + disk systems at $t < 1$ s), the dynamo cycle operates on timescales comparable to the orbital period, generating cyclically amplifying magnetic fields and periodic mass eruptions.

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2. System Parameters

- **Disk mass**: $M_{disk} = 0.03 M_\odot$
- **Central BH**: $M_{BH} = 3 M_\odot$, spin $\chi = 0.8$
- **Simulation duration**: 380 ms
- **Neutrino treatment**: full spectral transport (12 energy bins)
- **Peak neutrino luminosity**: $L_\nu \approx 10^{52}$ erg/s (initial)

- **Average**: $L_\nu \approx 10^{51}$ erg/s (integrated over 380 ms)

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3. MRI Timescale

The Magneto-Rotational Instability (MRI) growth rate:

$$\tau_{MRI} = \frac{1}{\Omega}$$

where $\Omega = \left(\frac{GM_{BH}}{r^3}\right)^{1/2}$ is the local orbital frequency. At $r = 10GM/c^2$ for $M_{BH} = 3M_\odot$:

$$\tau_{MRI} \approx 1 \times 10^{-3} \text{ s} = 1 \text{ ms}$$

The dynamo cycle is approximately 20 MRI timescales:

$$\tau_{dynamo} \approx 20 \cdot \tau_{MRI} \approx 20 \text{ ms}$$

UQFF Layer 2:

$$g_{L2,MRI} = \frac{1}{\tau_{MRI}} \approx 10^3 \text{ m/s}^2$$

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4. Dynamo Magnetic Field Cycle

The dynamo cycle proceeds as: 1. **Amplification**: MRI amplifies toroidal B_ϕ from seed poloidal B_r 2. **Buoyancy**: Magnetic pressure gradient causes B_ϕ to rise to magnetic corona 3. **Reconnection**: Reconnection at $z \sim H_{disk}$ releases energy, seeds new poloidal B_r 4. **Reset**: Cycle repeats with $\tau_{dynamo} \approx 20$ ms

The cycle amplitude:

$$B_{max} \approx \sqrt{4\pi P_{gas}} \approx 10^{15} \text{ G at ISCO}$$

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5. Electron Fraction Self-Regulation

The electron fraction (neutron content) evolves under neutrino absorption:

$$\frac{dY_e}{dt} = \dot{Y}_{e,abs} + \dot{Y}_{e,emit}$$

Midplane equilibrium: $Y_e^{eq} \approx 0.1$ (strongly neutron-rich, r-process capable)

Outflow Y_e: rises to $□$0.2 as material is heated by neutrino absorption during ejection

$$g_{L3,Ye} = \frac{M_{ej} \cdot Y_e}{r} \approx 2 \times 10^{-7} \text{ m/s}^2$$

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6. Dynamo Correction to UQFF

The dynamo cycle enters Layer 4:

$$g_{L4,dynamo} = \frac{\tau_{dynamo}}{r} \approx 2 \times 10^{-6} \text{ m/s}^2$$

This represents the periodic magnetic pressure wave launched at each dynamo reset, propagating through the disk corona at Alfvén speed.

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7. Mass Ejection Budget

Total ejecta from 380 ms simulation: - $M_{ej} \approx 0.01 M_\odot$ ($\approx 40\%$ of initial M_{disk}) - $v_\infty \approx 0.1c$ (thermal wind component) - $v_{fast} \approx 0.3c$ (MRI-driven MHD winds) - Average $Y_e \approx 0.2$, containing both lanthanide-rich and lanthanide-poor ejecta

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8. GW170817 Red Kilonova Connection

The neutrino-cooled disk wind, with $Y_e \approx 0.2$ and $M_{ej} \approx 0.01 M_\odot$, matches the GW170817 red kilonova: - Kilonova peak time: $□$3–5 days at visual/NIR wavelengths - Opacity $\kappa \approx 5$–10 cm2/g (partial lanthanide enrichment) - Third-peak r-process elements (Xe, Ba, Nd) require $Y_e \leq 0.25$

The dynamo cycle creates periodic neutrino bursts that additionally heat the ejecta, producing optical pulsations at $\tau_{dynamo} \approx 20$ ms (not yet observed but predicted).

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9. Summary

The 3D neutrino-cooled disk simulates a 20 ms magnetic dynamo cycle driven by MRI at $\Omega^{-1} \approx \tau_{MRI}$. Electron fraction self-regulates from $Y_e \approx 0.1$ midplane to $Y_e \approx 0.2$ outflow. Mass ejection $M_{ej} \approx 0.01 M_\odot$ at $v_\infty \approx 0.1c$ matches GW170817 red kilonova. These parameters are now registered in the Quadriadic UQFF Layers 2–4.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **BH-gravity** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u\phi_{\text{BH}})(\partial^\mu \phi_{\text{BH}}) - V(\phi_{\text{BH}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,[SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{BH}}) = \frac{1}{2}m^2\phi_{\text{BH}}^2 + \frac{\lambda}{4!}\phi_{\text{BH}}^4 + \kappa \cdot \rho_{\text{vac,[SCm]}} \cdot \phi_{\text{BH}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{BH}}} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R + \rho_{\text{vac,[SCm]}}g_{\mu\nu} + F_{U\_Bi\_i}/r^2 = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{\text{4 forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S/\delta\phi_{\text{BH}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac,[SCm]}}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac,[SCm]}} \cdot \text{exp!}\left(-\,\text{exp!}\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.173 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum.

This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 31, \quad n_{\text{channel}} = 15/26$$

Since $p_{\text{DVP}} = 31$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 protonuclear shell formation: the DPM proportion pair $(f'_{\text{UA}} + f_{\text{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **106 M_BH/M_M_sun yr** (quasi-normal mode ringdown):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_{\odot}}\right) \cdot \text{cos!} \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \text{tanh!} \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.173	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\text{DVP}} = 31$	PASS Resonant
BSH layers	26 harmonic terms	j = 1...26, cos$(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1×10-52 m-2 (UQFF vacuum term)	1.114×10-52 m-2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day → Γ_p suppression	< 4.17×10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_U_Bi_i` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n * sigma_n^SCm(omega) * Phi_phonon * (F_{U,Bi}/F_U - 1) where sigma_n^SCm(omega,n) = sigma_0 * exp[-(omega-omega_SCm)$^{2/(2*\mathrm{Gamma}}$2)] * (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + $2^{\{1-26\}/(1-2}${1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - f_26(q) = Sum_{n=0}^{25} $q^{\{n}$2} / (-q;q)*n^2 - phi_26(q) = Sum*{n=0}^{25} $q^{\{n}$2} / (-$q^{2;q}$2)*n - psi_26(q) = Sum*{n=1}^{26} $q^{\{n}$2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n * (1103+26390n) * W_26(n) / C_26 where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance

Symbol	Value	Description
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_1_CoAnQi.cpp`, and Wolfram kernels (`uqff_kozima_kernel.wl`, `uqff_s26_kernel.wl`, `uqff_mock_theta_pi_kernel.wl`).